

▼ Gutenberg Corpus

```
import nltk
nltk.download('gutenberg')
```

```
↳ [nltk_data] Downloading package gutenberg to /root/nltk_data...
[nltk_data] Unzipping corpora/gutenberg.zip.
True
```

```
['austen-emma.txt', 'austen-persuasion.txt', 'austen-sense.txt', 'bible-kjv.txt',
'blake-poems.txt', 'bryant-stories.txt', 'burgess-busterbrown.txt',
'carroll-alice.txt', 'chesterton-ball.txt', 'chesterton-brown.txt',
'chesterton-thursday.txt', 'edgeworth-parents.txt', 'melville-moby_dick.txt',
'milton-paradise.txt', 'shakespeare-caesar.txt', 'shakespeare-hamlet.txt',
'shakespeare-macbeth.txt', 'whitman-leaves.txt']
```

```
↳ ['austen-emma.txt',
'austen-persuasion.txt',
'austen-sense.txt',
'bible-kjv.txt',
'blake-poems.txt',
'bryant-stories.txt',
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'shakespeare-caesar.txt',
'shakespeare-hamlet.txt',
'shakespeare-macbeth.txt',
'whitman-leaves.txt']
```

```
emma = nltk.corpus.gutenberg.words('austen-emma.txt')
len(emma)
```

```
↳ 192427
```

```
emma_text = nltk.Text(gutenberg.words('austen-emma.txt'))
emma_text.concordance("surprize")
```

```
↳ Displaying 25 of 37 matches:
er father , was sometimes taken by surprize at his being still able to pity `
hem do the other any good ." " You surprize me ! Emma must do Harriet good : a
Knightley actually looked red with surprize and displeasure , as he stood up ,
r . Elton , and found to his great surprize , that Mr . Elton was actually on
d aid ." Emma saw Mrs . Weston ' s surprize , and felt that it must be great ,
father was quite taken up with the surprize of so sudden a journey , and his f
y , in all the favouring warmth of surprize and conjecture . She was , moreove
he appeared , to have her share of surprize , introduction , and pleasure . Th
ir plans ; and it was an agreeable surprize to her , therefore , to perceive t
talking aunt had taken me quite by surprize , it must have been the death of m
f all the dialogue which ensued of surprize , and inquiry , and congratulation
the present . They might chuse to surprize her ." Mrs . Cole had many to agre
the mode of it , the mystery , the surprize , is more like a young woman ' s s
to her song took her agreeably by surprize -- a second , slightly but correct
" " Oh ! no -- there is nothing to surprize one at all .-- A pretty fortune ;
t to be considered . Emma ' s only surprize was that Jane Fairfax should accep
of your admiration may take you by surprize some day or other ." Mr . Knightle
ation for her will ever take me by surprize .-- I never had a thought of her i
expected by the best judges , for surprize -- but there was great joy . Mr .
sound of at first , without great surprize . " So unreasonably early !" she w
d Frank Churchill , with a look of surprize and displeasure .-- " That is easy
; and Emma could imagine with what surprize and mortification she must be retu
tled that Jane should go . Quite a surprize to me ! I had not the least idea !
. It is impossible to express our surprize . He came to speak to his father o
g engaged !" Emma even jumped with surprize ;-- and , horror - struck , exclai
```

```
from nltk.corpus import gutenberg
gutenberg.fileids()
['austen-emma.txt', 'austen-persuasion.txt', 'austen-sense.txt', ...]
emma = gutenberg.words('austen-emma.txt')
```

```
import nltk
nltk.download('punkt_tab')
```

```

[ntlk_data] Downloading package punkt_tab to /root/nltk_data...
[ntlk_data] Unzipping tokenizers/punkt_tab.zip.
True

```

```

for fileid in gutenbergl.fileids():
    num_chars = len(gutenberg.raw(fileid))
    num_words = len(gutenberg.words(fileid))
    num_sents = len(gutenberg.sents(fileid))
    num_vocab = len(set(w.lower() for w in gutenberg.words(fileid)))
    print(round(num_chars/num_words), round(num_words/num_sents),
          round(num_words/num_vocab), fileid)

```

```

5 25 26 austen-emma.txt
5 26 17 austen-persuasion.txt
5 28 22 austen-sense.txt
4 34 79 bible-kjv.txt
5 19 5 blake-poems.txt
4 19 14 bryant-stories.txt
4 18 12 burgess-busterbrown.txt
4 20 13 carroll-alice.txt
5 20 12 chesterton-ball.txt
5 23 11 chesterton-brown.txt
5 18 11 chesterton-thursday.txt
4 21 25 edgeworth-parents.txt
5 26 15 melville-moby_dick.txt
5 52 11 milton-paradise.txt
4 12 9 shakespeare-caesar.txt
4 12 8 shakespeare-hamlet.txt
4 12 7 shakespeare-macbeth.txt
5 36 12 whitman-leaves.txt

```

```

raw = gutenberg.raw("burgess-busterbrown.txt")
print(raw[1:20])

```

```

words = gutenberg.words("burgess-busterbrown.txt")
print(words[1:20])

```

```

sents = gutenberg.sents("burgess-busterbrown.txt")
print(sents[1:20])

```

```

The Adventures of B
['The', 'Adventures', 'of', 'Buster', 'Bear', 'by', 'Thornton', 'W', '.', 'Burgess', '1920', ''], 'I', 'BUSTER', 'BEAR', 'GOES', 'F]
[['I'], ['BUSTER', 'BEAR', 'GOES', 'FISHING'], ['Buster', 'Bear', 'yawned', 'as', 'he', 'lay', 'on', 'his', 'comfortable', 'bed', 'c

```

```

macbeth_sentences = gutenberg.sents('shakespeare-macbeth.txt')
macbeth_sentences
[['The', 'Tragedie', 'of', 'Macbeth', 'by', 'William', 'Shakespeare',
'1603', ''], ['Actus', 'Primus', '.'], ...]
macbeth_sentences[1116]
['Double', ',', 'double', ',', 'toile', 'and', 'trouble', ';',
'Fire', 'burne', ',', 'and', 'Cauldron', 'bubble']
longest_len = max(len(s) for s in macbeth_sentences)
[s for s in macbeth_sentences if len(s) == longest_len]
[['Doubtfull', 'it', 'stood', ',', 'As', 'two', 'spent', 'Swimmers', ',', 'that',
'doe', 'cling', 'together', ',', 'And', 'choake', 'their', 'Art', ':', 'The',
'mercilesse', 'Macdonwald', ...]]

```

```

[['Doubtfull',
'it',
'stood',
',',
'As',
'two',
'spent',
'Swimmers',
',',
'that',
'doe',
'cling',
'together',
',',
'And',
'choake',
'their',
'Art',
':',
'The',
'mercilesse',
'Macdonwald',
Ellipsis]]

```

✓ Brown Corpus

```
import nltk
nltk.download('brown')
```

```
↳ [nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Unzipping corpora/brown.zip.
True
```

```
from nltk.corpus import brown
brown.categories()
```

```
brown.words(categories='news')
```

```
brown.words(fileids=['cg22'])
```

```
brown.sents(categories=['news', 'editorial', 'reviews'])
```

```
↳ [['The', 'Fulton', 'County', 'Grand', 'Jury', 'said', 'Friday', 'an', 'investigation', 'of', "Atlanta's", 'recent', 'primary',
'election', 'produced', '', 'no', 'evidence', "", 'that', 'any', 'irregularities', 'took', 'place', '.'], ['The', 'jury',
'further', 'said', 'in', 'term-end', 'presentments', 'that', 'the', 'City', 'Executive', 'Committee', ',', 'which', 'had', 'over-
all', 'charge', 'of', 'the', 'election', ',', '', 'deserves', 'the', 'praise', 'and', 'thanks', 'of', 'the', 'City', 'of',
'Atlanta', "", 'for', 'the', 'manner', 'in', 'which', 'the', 'election', 'was', 'conducted', '.'], ...]
```

```
from nltk.corpus import brown
news_text = brown.words(categories='news')
fdist = nltk.FreqDist(w.lower() for w in news_text)
modals = ['can', 'could', 'may', 'might', 'must', 'will']
for m in modals:
    print(m + ':', fdist[m], end=' ')
```

```
↳ can: 94 could: 87 may: 93 might: 38 must: 53 will: 389
```

```
cfd = nltk.ConditionalFreqDist(
(genre, word)
for genre in brown.categories()
for word in brown.words(categories=genre))
genres = ['news', 'religion', 'hobbies', 'science_fiction', 'romance', 'humor']
modals = ['can', 'could', 'may', 'might', 'must', 'will']
cfd.tabulate(conditions=genres, samples=modals)
```

```
↳
```

	can	could	may	might	must	will
news	93	86	66	38	50	389
religion	82	59	78	12	54	71
hobbies	268	58	131	22	83	264
science_fiction	16	49	4	12	8	16
romance	74	193	11	51	45	43
humor	16	30	8	8	9	13

Start coding or [generate](#) with AI.